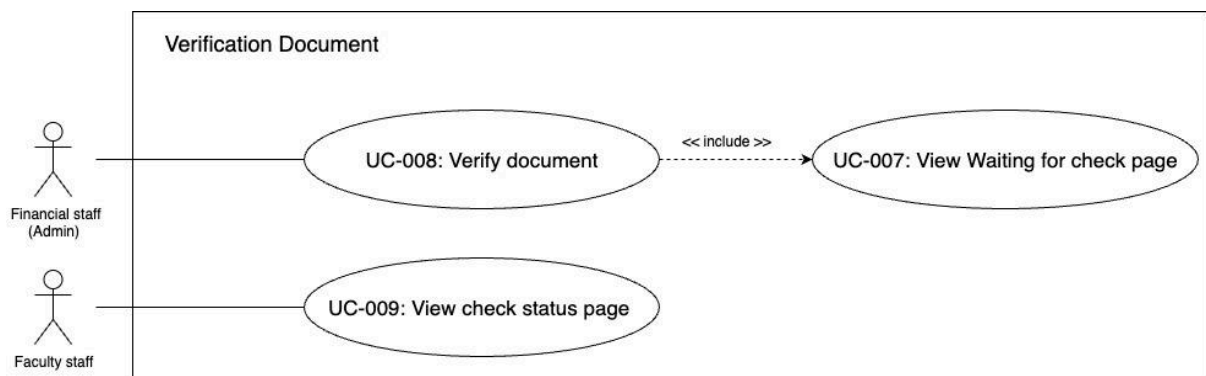


Use case: Admin Verification Document

This feature allows financial staff as administrators to verify documents submitted by faculty staff by comparing them with the original reference documents. The system extracts data from both the user-uploaded and admin-uploaded files, then presents them side by side for easy verification. Key fields such as invoice numbers, doc types, supplier names, amounts, payment dates, and signatures, invoice numbers, doc types, supplier names, amounts, payment dates, and signature must match before approval is granted. If it does not match, the admin can reject the submission and request a correction from the faculty staff by indicator status (e.g., Approved, Pending, Rejected), making it easier to manage document workflows and monitor progress.

Usecase Diagram



Activity Diagram

UseCase Description : View Waiting for check page

Use case ID	UC-007	
Use case name	View Waiting for check page	
Actors	Financial Staff (Admin)	
Description	Admin can view a list of financial documents submitted by staff and can search for documents using the search bar and date range filter functions.	
Trigger	The process begins when staff send financial documents to the admin.	
Preconditions	Financial staff must be logged into the system and must be authenticated as an admin.	
Use Case Input Specification		
Input	Type	Constraint
User ID	String	The ID of the currently signed-in user, retrieved from the user's session.
Post Conditions	The system will display a list of financial documents submitted by staff but not yet reviewed.	
Normal Flows		
User	System	
1. The admin opens “Waiting for check” page		
	2. The system retrieves a list of financial documents submitted by staff from the database, sorted by date and time submitted. [E1: Server error]	
	3. The system displays a search box, filter function and a list of financial documents submitted by staff. [A1: Have a search term in the search box] [A2: Filter function select a time period]	
	4. A list of financial documents submitted by staff including the document name, title, sender, department, date and time submitted by the staff.	
Alternative Flow	[A1: Have a search term in the search box] A1.1: The system will query financial documents submitted by staff. [A2: Filter function select a time period]	

	• A2.1: The system displays financial documents submitted by staff that found that time period.
Exception Flow	[E1: Database Connection Error] • E1.1: The system cannot establish a connection with the database. • E1.2: The system will display an alert: "Error: Unable to retrieve document list from database. Please try again later" [E2: Internal Server Error] • E2.1: The system encounters an unexpected error while querying for the document list. • E2.2: The system will display an alert: "Error: An unexpected server error occurred while retrieving the document list. Please try again later."

URS-007: The admin clicks at the “Waiting for check” menu to view a list of financial documents submitted by staff.

SRS-028: The system retrieves a list of financial documents submitted by staff from the database, sorted by date and time submitted.

SRS-029: The “Waiting for check” page must display a search box, date range filter function and a list of financial documents submitted by staff.

SRS-030: The system does not have a search term in the search box, the system displays all lists of financial documents submitted by staff.

SRS-031: The system has a search box with search terms, the system will query financial documents submitted by staff.

SRS-032: The date range filter function must select a time period to display financial documents submitted by staff that found that time period.

SRS-033: The system must display a list of financial documents submitted by staff including the document name, title, sender, department, date and time submitted by the staff.

UseCase Description : Verify document

Use case ID	UC-008		
Use case name	Verify document		
Actors	Financial Staff (Admin)		
Description	Admin can view the financial documents submitted by staff to check whether the values in the reference document and the financial document match and return the document status to staff.		
Trigger	The process begins when staff send financial documents to the admin.		
Preconditions	The admin on the waiting for check page.		
Use Case Input Specification			
Input	Type	Constraint	Example
document_id	string	Must reference a valid pending document uploaded by staff.	2
decision	enum	Values = {Approve, Reject}	“Approve”
reject_reason	string	Mandatory if decision = Rejected, must not be empty.	"Signature missing"
User		System	
1. The admin clicks on financial documents submitted by staff in the “Waiting for check” page.			
		2. The system retrieves the reference documents and financial documents from staff are automatically retrieved from the database.	
		3. The system on the left side displays preview file area, table of extraction data of a reference document such as document type, invoice number, supplier name, date, amount and signature while on right side displays preview file area, table of the extraction data of financial documents from staff such as document type, invoice number, supplier name, date, amount and signature to check information along with the approve or reject this document option and the confirm and cancel buttons. [E1: Server error]	
		4. The values such as amount and signature from both the extracted data from the reference document and the extracted data from the financial document will match on both sides. The text turns green to indicate that the data on both sides match. [A1: Value is not matched]	

5. The admin selects the “Approved” status and click “Confirm” button in reject reason popup [A2: The admin selects the “Rejected” status and click “Confirm” button] [A3: The admin click “Cancel” button]	
	7. The system will return the status to the staff.
	8. The system displays an alert message “The document has been successfully checked.”
Alternative Flow	[A1: Value is not matched] • A1.1 The system will turn the text to red to indicate that the information on both sides does not match. [A2: The admin selects the “Rejected” status and click “Confirm” button] • A2.1 The system will popup the “Reject reason” window • A2.2 The “Reject reason” popup window displays a text input field with “Submit” and “Cancel” button. • A2.3 The admin clicks “Submit” button in reject reason popup [A4: The admin click “Cancel” in reject reason popup] • A2.4 The system will return the status to the staff. • A2.5 The system displays an alert message “The document has been successfully checked.” [A3: The admin click “Cancel” button] • A3.1 The system returns to the “Waiting for check” page. [A4: The admin click “Cancel” in reject reason popup] • A4.1 The system will close this popup window.
Exception Flow	[E1: Database Connection Error] • E1.1: The system cannot establish a connection with the database. • E1.2: The system will display an alert: "Error: Unable to retrieve document data from database. Please try again later" [E2: Internal Server Error] • E2.1: The system encounters an unexpected error while querying for the document list. • E2.2: The system will display an alert: "Error: An unexpected server error occurred while retrieving the document list. Please try again later."

URS-013: The admin clicks on a financial document submitted by staff to verify the document by comparing the reference document with the financial document.

SRS-034: Upon the admin click on a financial document submitted by staff at Waiting for check page that navigate to Verification document page.

SRS-035: The system retrieves the reference documents and financial documents from staff are automatically retrieved from the database.

SRS-036: The “Verification document” page on left side displays preview file area, table of extraction data of a reference document such as document type, invoice number, supplier name, date, amount and signature while on right side displays preview file area, table of the extraction data of financial documents from staff such as document type, invoice number, supplier name, date, amount and signature to check information along with the approve or reject this document option and the “Confirm” and “Cancel” buttons.

SRS-037: When the values such as amount and signature from both the extracted data from the reference document and the extracted data from the financial document will match on both sides. The text turns green to indicate that the data on both sides match.

SRS-038: When the values such as amount and signature from both the extracted data from the reference document and the extracted data from the financial document do not match. The text turns red to indicate that the data on both sides do not match.

SRS-039: Upon the admin click “Approved” status and then clicks on the Confirm button, the system will return the status to the staff and display an alert message “The document has been successfully checked.”

SRS-040: Upon the admin click “Rejected” status and then clicks on the Confirm button, the system will popup the “Reject reason” window.

SRS-041: The reject reason popup window displays a text input field with "Submit" and "Cancel" button.

SRS-042: Upon the admin click the "Submit" button in the "Reject reason" popup window, the system will return the status to the staff and display an alert message "The document has been successfully checked."

SRS-043: Upon the admin click the "Cancel" button in the "Reject reason" popup window, the system will close this popup window.

SRS-044: Upon the admin click the "Cancel" button returns to the "Waiting for check" page.

UseCase Description : View Check status page

Use case ID	UC-009		
Use case name	View Check status page		
Actors	Faculty staff		
Description	Staff can view lists of submitted financial documents and can search for documents using the search bar,status filter function and date range filter functions.		
Trigger	The process begins when sending financial documents to the admin.		
Preconditions	The staff confirmed the extracted financial data already.		
Use Case Input Specification			
Input	Type	Constraint	Example
search_file	string	Optional; may be empty.	“invoicetax”
status_filter	string	Values = {Pending, Approved, Rejected}	“Approve”
date_range (from_date, to_date)	datetime	must follow a valid date format.	2025-07-01 to 2025-07-31
Post Conditions	The system keeps a record of documents that staff have sent to the admin and have a pending, approved, or rejected status.		
Normal Flows			
User		System	
1. The staff opens “Check Status” page			
		2. The system retrieves all lists of submitted financial documents that have been sent to the admin, sorted by date and time of submission. [E1: Database Connection Error] [E2: Internal Server Error]	
		3. The system displays a search box, status filter function, date range filter function and a list of submitted financial documents. [A1: Have a search term in the search box] [A2: Filter function select a time period]	
		4. A list of financial documents submitted to the admin for review, including the document name, title, date and time of submission, and document status such as pending,	

	approved, or rejected.
5. The staff click on a list of financial documents submitted.	
	6. A modal will appear with the document name, title, status, date and time of submission, file url, extracted financial data table and close button. [A3: The staff click the "Close" button in modal]
Alternative Flow	[A1: Have a search term in the search box] - A1.1: The system will query financial documents submitted. [A2: Filter function select a time period] - A2.1: The system displays financial documents submitted that found that time period. [A3: The staff click the "Close" button in modal] - A3.1: The system will close the modal.
Exception Flow	[E1: Database Connection Error] - E1.1: The system cannot establish a connection with the database. - E1.2: The system will display an alert: "Error: Unable to retrieve document list from database. Please try again later" [E2: Internal Server Error] - E2.1: The system encounters an unexpected error while querying for the document list. - E2.2: The system will display an alert: "Error: An unexpected server error occurred while retrieving the document list. Please try again later."

URS-014: Staff clicks at the "Check Status" menu to view a list of financial documents that have been sent to the admin for review.

SRS-045: The system retrieves all lists of submitted financial documents that have been sent to the admin, sorted by most recent submission date and time.

SRS-046: The "Check Status" page must display a search box, status filter function, date range filter function and a list of submitted financial documents.

SRS-047: The system does not have a search term in the search box, the system displays all lists of submitted financial documents.

SRS-048: The system has a search box with search terms, the system will query submitted financial documents.

SRS-049: The date range filter function must select a time period to display a submitted financial document that found that time period.

SRS-050: The status filter function must select a status to display a submitted financial document based on their status such as pending, approved, rejected.

SRS-051: The system must display a list of financial documents submitted to the admin for review, including the document name, title, date and time of submission, and document status such as pending, approved, or rejected.

SRS-052: Upon the staff click on the financial documents submitted a modal will appear with the document name, title, status, date and time of submission, file url, extracted financial data table and close button.

- If the document status is rejected, a rejected reason from staff will be displayed.

SRS-053: Upon the staff click the "Close" button in modal, the modal is closed.

Method declaration

Method ID: M1

Method Name: listPendingUploads()

Description:

Retrieves all documents currently in Pending status from uploadfiles for the admin verification dashboard. Requires an authenticated Admin access token. Returns the operation status and a collection of pending uploads, each containing core file and uploader details for filtering and review.

Input Parameters:

Parameter	Type	Description
auth_user_id	string	The user_id of the authenticated Admin caller. Value comes from the user's session or token and must correspond to a user with an "Admin" role.

(This value comes from the user's session or token and must correspond to a user whose role = "Admin.")

Return:

Field	Type	Description
status	string	"success" =The document list was retrieved successfully. "error" =An error occurred. the documents array will not be returned.
documents	array	An array of document objects, each containing key information such as document_id, file_name, submitter_name, departments, submission_date, status.

-for a Database Connection Error:

```
{ "status": "error", "message": "Error: Unable to retrieve document list from database. Please try again later" }
```

-for an Internal Server Error:

```
{ "status": "error", "message": "Error: An unexpected server error occurred while retrieving the document list. Please try again later." }
```

Method ID: M2

Method Name: getExtractedByFile(fileId)

Description:

Fetches the extracted data rows for a selected pending document so an Admin can verify and confirm. Requires Admin token. Returns the operation status and a list of item rows containing document metadata and extracted financial fields.

Input Parameters:

Parameters	Type	Description
auth_admin_id	string	Identifier of the logged-in Admin performing the verification.
file_id	string	The target file's identifier.

Return:

Field	Type	Description
status	string	" success " = if details are available. " error " = Load data details failed.
data_detail	string	data details: page, doc_type, doc_type_id, bill_number, payment_date, amount, signature.

-for a Database Connection Error:

```
{ "status": "error", "message": "Error: Unable to retrieve document list from database.  
Please try again later" }
```

-for an Internal Server Error:

```
{ "status": "error", "message": "Error: An unexpected server error occurred while  
retrieving the document list. Please try again later." }
```

Method ID: M3

Method Name: approveUpload(file_id)

Description:

Marks a selected pending document as Approved. Requires Admin token. On success, the document's status is updated to "approved". Returns the operation status and a confirmation message. The client may trigger a refresh of the pending list after completion.

Input Parameters:

Parameter	Type	Description
auth_user_id	string	The user_id of the user to be deleted.
document_id	string	The ID of the document to be approved.

Return:

Field	Type	Description
status	string	"success" =Document was approved successfully. "error" =An error occurred.
message	string	Provides confirmation on success or an explanation of the error.

-If the process is successful, the system will return:

```
{ "status": "success", "message": "Document [document_id] approved successfully." }
```

-If a database error occurs while saving, show:

```
{ "status": "error", "message": "Failed to update document status due to a database error." }
```

Method ID: M4

Method Name: rejectUpload(file_id, reason)

Description:

Rejects a document after failed verification. This method updates the document's status to "Rejected," saves the reason, and triggers a rejection notification.

Input Parameters:

Parameter	Type	Description
auth_user_id	string	The user_id of the user to be deleted.
document_id	string	The ID of the document to be rejected.
reason	string	The text description explains why the document was rejected.

Return:

Field	Type	Description
status	string	"success" =Document was approved successfully. "error" =An error occurred.
message	string	Provides confirmation on success or an explanation of the error.

-If the reason field is empty, show:

```
{ "status": "error", "message": "Rejection reason cannot be empty." }
```

-if the system error while reject, show:

```
{ "status": "error", "message": "Reject failed." }
```

-If the process is successful, the system will return:

```
{ "status": "success", "message": "Document [document_id] approved successfully." }
```

-If a database error occurs while saving, show:

```
{ "status": "error", "message": "Failed to update document status due to a database error." }
```

Unit testing

Test Data

Test data	Data
UTC-M1-TD-01	auth_user_id = "admin_01", file_id = "1,2"
UTC-M1-TD-02	auth_user_id = "user_02"
UTC-M1-TD-03	auth_user_id = "admin_01"
UTC-M1-TD-04	auth_user_id = "admin_02"

Unit Testing: M1 - listPendingUploads()

Test ID	Scenario	Input	Expected result
UT-M1-01	Successful Retrieval: An authorized Admin retrieves the list of pending documents.	auth_user_id: "admin_01"	Returns { "status": "success", "documents": [{เอกสารใบขับขี่}, {ใบยื่นเบิกเงิน}] }
UT-M1-02	Unauthorized Access: A non-Admin user attempts to retrieve the list.	auth_user_id: "user_02"	Returns { "status": "error", "message": "Unauthorized access." }
UT-M1-03	Database Error: An error occurs while fetching from the database.(Mock up by changing data base password)	auth_user_id: ["admin_01"]	Returns {"status":"error","message": "Unable to retrieve document list from database. Please try again later."}
UT-M1-04	No pending documents	auth_user_id: "admin_02"	Returns {"status":"success","documents":[] }

Test Data

Test data	Data
UTC-M2TD-01	auth_admin_id = "admin_01", file_id = "doc_1"
UTC-M2-TD-02	auth_admin_id = "admin_01", file_id = "doc_999"
UTC-M2-TD-03	auth_admin_id = "user_02"
UTC-M2-TD-04	auth_admin_id = "admin_01", file_id = "doc_2"

Unit Testing: M2 - getExtractedByFile(fileId)

Test ID	Scenario	Input	Expected result
UT-M2-01	Successful Upload: Admin requests detail of a specific document that exists. The system must return all extracted fields.	auth_admin_id:"admin_01", file_id:"doc_1"	Returns { "status": "success", "data_detail": { "page": 1, "doc_type": "ใบดิ่งหนี", "bill_number": "INV-7012", "payment_date": "25/08/67", "amount": 50,000, "signature": "คมสันต์"} }
UT-M2-02	Document Not Found: Admin requests details of a document ID that does not exist. The system must return an error message.	auth_admin_id:"admin_01", file_id:"doc_999"	Returns { "status": "error", "message": "Unable to load extracted data." }
UT-M2-03	Unauthorized Access: A non-Admin user tries to load document details. The system must return an unauthorized error.exceeds the 25MB size limit.	auth_admin_id:"user_02", file_id:"doc_1"	Returns { "status": "error", "message": "Unauthorized access." }
UT-M2-04	Database Error: A database failure happens while loading the document detail. The system must return a generic error message.(Mock up by changing the database password)	auth_admin_id:"admin_01", file_id:"doc_2"	Returns { "status": "error", "message": "Unable to load extracted data." }

Test Data

Test data	Data
UTC-M3-TD-01	auth_user_id = "admin_01", document_id = "doc_1"
UTC-M3-TD-02	auth_user_id = "admin_01", file_id = "doc_999"
UTC-M3-TD-03	auth_user_id = "admin_01", document_id = "doc_2"
UTC-M3-TD-04	auth_user_id = "admin_01", file_id = "doc_3"
UTC-M3-TD-05	auth_user_id = "user_02", document_id = "doc_1"

Unit Testing: M3 - approveUpload(file_id)

Test ID	Scenario	Input	Expected result
UT-M3-01	Successful Approval: An Admin approves a pending document.	auth_user_id: "admin_01", document_id: "doc_1"	Returns { "status": "success", "message": "Document doc_1 approved successfully." }
UT-M3-02	Document Not Found: The provided document_id does not exist.	auth_user_id: "admin_01", document_id: "doc_999"	Returns { "status": "error", "message": "Document not found or is not in a pending state." }
UT-M3-03	Document Not Pending: An attempt is made to approve a document that is already processed (e.g., 'Approved' or 'Rejected').	auth_user_id: "admin_01", document_id: "doc_2"	Returns { "status": "error", "message": "Document not found or is not in a pending state." }
UT-M3-04	Database Error: An error occurs while saving the new status.(Mock up by changing database password)	auth_user_id: "admin_01", document_id: "doc_3"	Returns { "status": "error", "message": "Failed to update document status due to a database error." }
UT-M3-05	Unauthorized (non-admin): non-Admin user attempts to approve a document.	auth_user_id:"user_02", document_id:"doc_1"	Returns { "status": "error", "message": "Una uthorized access." }

Test Data

Test data	Data
UTC-M4-TD-01	auth_user_id = "admin_01", document_id = "doc_1", reason = "Signature mismatch."
UTC-M4-TD-02	auth_user_id = "admin_01", document_id = "doc_2", reason = ""
UTC-M4-TD-03	auth_user_id = "admin_01", document_id = "doc_111"
UTC-M4-TD-04	auth_user_id = "admin_01", document_id = "doc_456", reason = "Already processed"
UTC-M4-TD-05	auth_user_id = "user_02", document_id = "doc_1", reason = "N/A"
UTC-M4-TD-06	auth_user_id = "admin_01", document_id = "doc_3", reason = "Mismatch"

Unit Testing: M4 - rejectUpload(file_id, reason)

Test ID	Scenario	Input	Expected result
UT-M4-01	Successful Rejection: An Admin rejects a pending document with a valid reason.	auth_admin_id: "admin_01", document_id: "doc_1", reason: "Signature mismatch."	Returns { "status": "success", "message": "Document doc_1 rejected successfully." }
UT-M4-02	Reason Missing: An attempt to reject without providing a reason.	auth_admin_id: "admin_01", document_id: "doc_2", reason: ""	Returns { "status": "error", "message": "Rejection reason cannot be empty." }
UT-M4-03	Document Not Found: The provided document_id does not exist.	auth_admin_id: "admin_01", document_id: "doc_111", reason: "Test"	Returns { "status": "error", "message": "Document not found or is not in a pending state." }
UT-M4-04	Document Not Pending: An attempt is made to reject a document that is already processed.	auth_admin_id: "admin_01", document_id: "doc_456", reason: "Test"	Returns { "status": "error", "message": "Document not found or is not in a pending state." }
UT-M4-05	Unauthorized (non-admin): A non-Admin user attempts to reject a document.	auth_user_id:"user_02", document_id:"doc_1", reason:"Am not a admin"	Returns { "status": "error", "message": "Unauthorized access." }

UT-M4-06	Database Error: A database failure occurs while rejecting a document. The system must return a failure message.(Mock up by changing data base password)	auth_admin_id:"admin_01", document_id:"doc_3", reason:"I don't know"	Returns { "status": "error", "message": "Reject failed." }
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UAT Testing

UAT-007: View “Waiting for check” page (Admin)

:Verify that an Admin can open the “Waiting for check” page and see pending uploads with search and date-range filtering.

Preconditions:

- Tester is logged in with an account that has **Admin** privileges.
- There are at least two **pending documents** with different submission dates and departments.

Test steps:

1. Navigate to **Waiting for check**.
2. Observe that the page loads a list of pending uploads.
3. Enter a **search term** that matches one document.
4. Apply a **date-range filter**.

Expected Result:

- The page displays a search box, date-range filter, and a list with document name, title, sender, department, and submission date/time.
- Without search/filter → all pending items are listed.
- With search → only matching items are listed.
- With date-range → only items within the range are listed.
- If no matches → an empty list is displayed (no crash).

UAT-008: Approve document (Admin)

:Verify that an Admin can approve a pending document after verifying extracted data against the reference.

Preconditions:

- Tester is logged in as **Admin**.
- At least one pending document exists with stored extracted data and a corresponding reference.

Test steps:

1. On **Waiting for check**, click a pending document → open **Verification page**.
2. Verify that **reference** and **staff-uploaded** documents are displayed side-by-side.
3. Check that matched fields are shown in **green**.
4. Select **Approved**, then click **Confirm**.

Expected Result:

- The system retrieves reference and staff documents automatically.
- Two panels show previews and tables with fields (doc type, invoice/bill no., supplier, amount, payment date, signature).
- Matching values are highlighted in green.
- On confirmation, status is updated to **Approved**, saved with approver info, and staff is notified.
- Alert is shown: "**The document has been successfully checked.**"

UAT-009: Cancel verification actions (Admin)

:Verify that Cancel actions work on the verification page and in the reject reason popup.

Preconditions:

- Tester is logged in as Admin.
- At least one pending document exists and is open in the Verification page.

Test steps:

1. Click Cancel on the Verification page.
2. Select Reject, click Confirm to open the popup, then click Cancel in the popup.

Expected Result:

- Step 1 → system returns to the Waiting for check page.
- Step 2 → reject the reason popup closes with no status change.

UAT-010: Reject document with reason (Admin)

:Verify that an Admin can reject a pending document with a reason when extracted values do not match.

Preconditions:

- Tester is logged in as Admin.
- At least one pending document exists where some extracted fields do not match the reference.

Test steps:

1. On the Verification page, observe non-matched fields highlighted in red.
2. Select Rejected, then click Confirm.
3. In the popup, enter a valid reason and click Submit.

Expected Result:

- Non-matched fields are highlighted in red.
- A Reject reason popup appears with text input.
- On Submit → status updated to Rejected, rejection reason recorded, staff notified.
- Alert is shown: "The document has been successfully checked."

UAT-011: View “Check Status” page (Staff)

:Verify that a Staff user can view all submitted documents with filters for search, status, and date-range.

Preconditions:

- Tester is logged in with a **Staff** account.
- Multiple submitted documents exist with different statuses and submission dates.

Test steps:

1. Navigate to **Check Status** page.
2. Observe the list sorted by **most recent submission time**.
3. Enter a search term.
4. Apply a **status filter** (e.g., Approved).
5. Apply a **date-range filter**.

Expected Result:

- Page displays search box, status filter, date-range filter, and list of submissions.
- List contains document name, title, submission date/time, and status.
- Filters narrow the results correctly.
- If no matches, an empty state is displayed.

UI(Prototype)

UI-001: Verification Document page



ABC@cmu.ac.th

 DASHBOARD

 Waiting for check

 Document archive

Status





01052568.pdf waiting for check

Faculty of Engineering

11.02
01 May 2025

18042568.pdf waiting for check

Faculty of Humanities

13.00
19 April 2025

17042568.pdf waiting for check

Faculty of Science

14.34
18 May 2025

UI-002: Compare document page



ABC@cmu.ac.th

 DASHBOARD

 Waiting for check

 Document archive

Reference Document

PREVIEW ZONE

Financial Key

Doc Type	Invoice Number	Supplier Name	Date	Amount	Signature

Financial Document

PREVIEW ZONE

Financial Key

Doc Type	Invoice Number	Supplier Name	Date	Amount	Signature

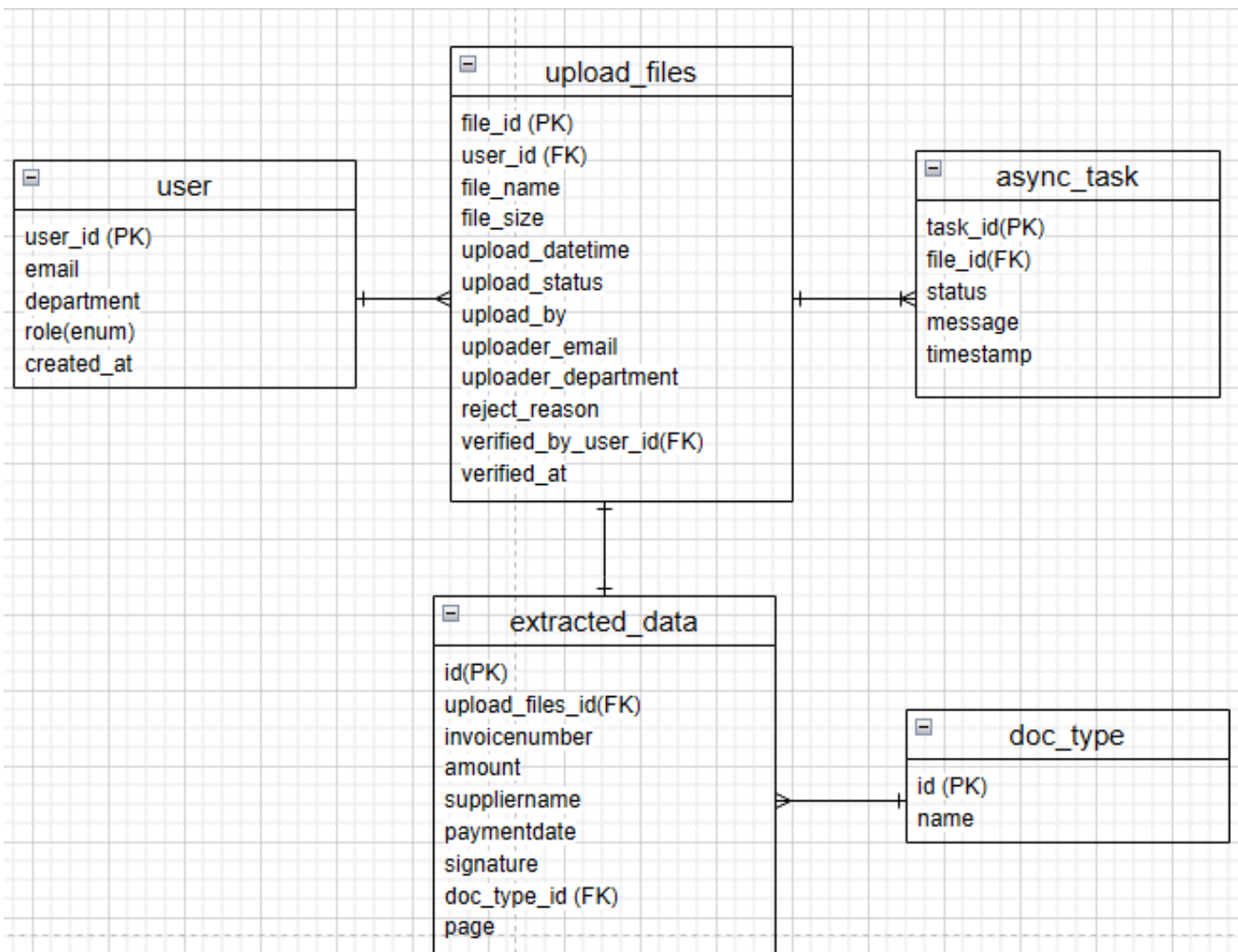
Approved

Rejected

CANCEL

CONFIRM

ER-Diagram



Data Structure

1. Table: User

Name	Type	Description	Example
user_id (PK)	int	A unique identifier for each user (Primary Key)	1
email	str	The user's email address, used for login and notifications.	chatchai.c@cmu.ac.th
department	str	The department or faculty the user belongs to	College of Arts, Media and Technology
role	str	The user's role in the system ('Faculty Staff', 'Financial Staff')	'Admin'
created_at	datetime	Date the assign	2025-06-23 10:30:00

2. Table: upload_files

Name	Type	Description	Example
file_id (PK)	int	Unique file identifier	1
user_id (FK)	str	Owner/uploader (ref user.user_id)	1
file_name	str	Original file name	invoice_001.pdf
file_size	str	File size	1.25MB
upload_datetime	datetime	Time uploaded	2025-08-01 12:34:56
upload_status	enum	pending / approved / rejected	pending
upload_by	str	Display name of uploader	Patara Nun
uploader_email	str	Uploader's email	patara_nun@cmu.ac.th
uploader_department	str	Uploader's department	College of art, media and technology
reject_reason	str	Reason when rejected (optional)	Missing signature
verified_by_user_id (FK)	int	Admin reviewer (ref user.user_id)	2
verified_at	datetime	Time of approve/reject	2025-08-02 09:45:10

3. Table: extracted_data

Name	Type	Description	Example
id (PK)	int	Unique file identifier	1
upload_files_id (FK)	int	Ref to upload_files.file_id	1
invoicenumber	str	Invoice/bill number	invoice_001
amount	str	Amount	15230.50
suppliername	str	Supplier name	ABC Supply
paymentdate	date	Payment/issued date	2025-07-31
signature	str	Signature text/marker	Patara Nun
doc_type_id (FK)	int	Ref to doc_type.id	1
page	int	Page number in the file	3

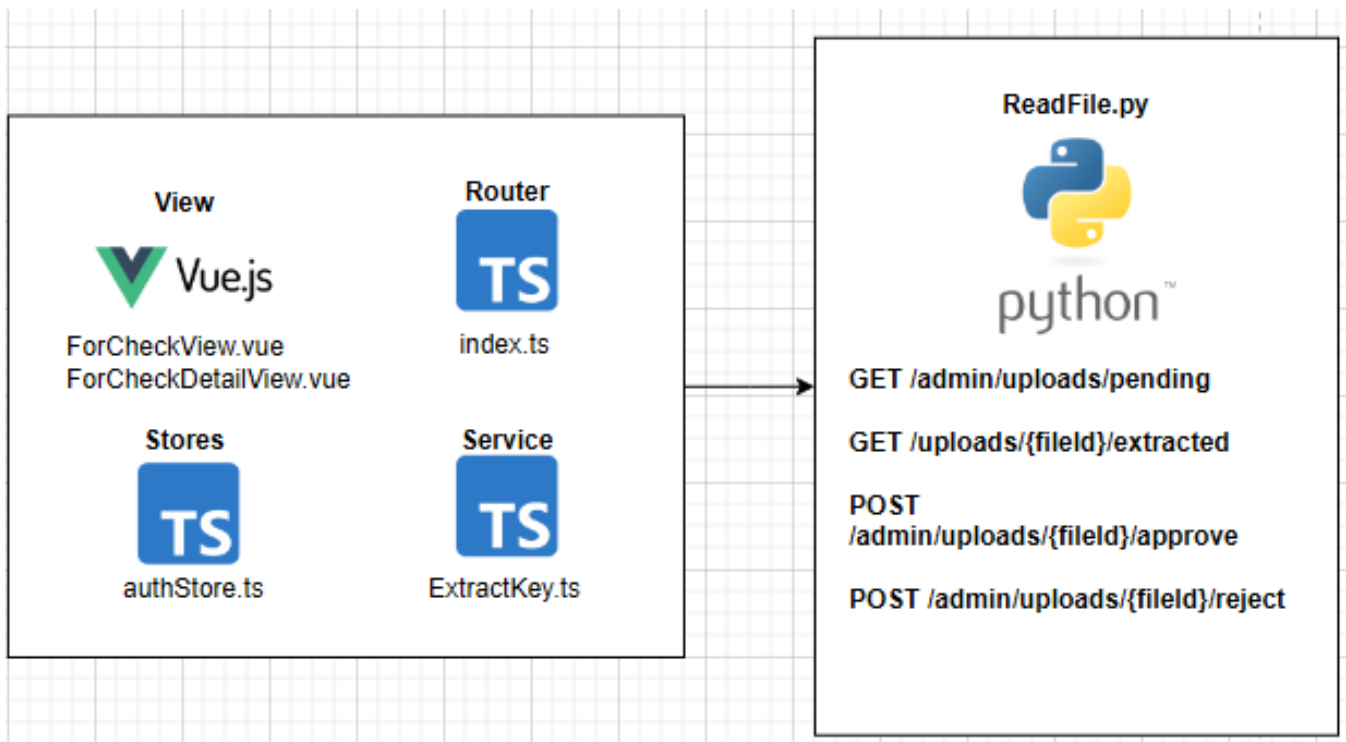
4. Table: doc_type

Name	Type	Description	Example
id (PK)	int	Document type id	1
name	str	Document type name	Tax Invoice

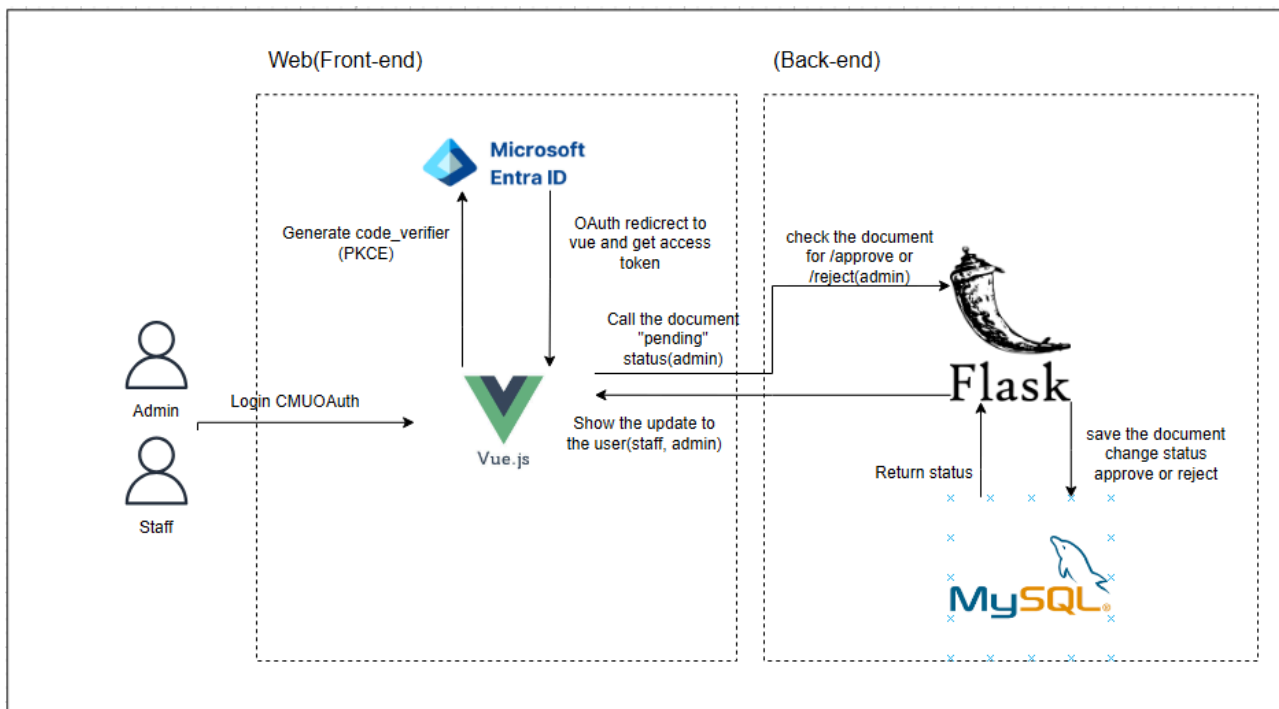
5. Table: async_task

Name	Type	Description	Example
task_id (PK)	str	Background job id for extraction	12b8f-...-f1c2
file_id (FK)	int	Ref to upload_files.file_id	101
status	enum	processing / complete / error	processing
message	str	Status message	OCR running...
timestamp	datetime	Status last updated	2025-08-01 12:36:10

Code Structure



System Architecture



The verification flow starts after an Admin has already signed in. From the Vue.js app, the Admin opens the verification page, which asks the Flask API for all files currently waiting for review. This is done with a secure request to the GET /admin/uploads/pending endpoint using the Admin's Bearer token, and the backend returns a list of pending uploads together with who submitted them and their department.

When the Admin clicks a specific file, the app requests the pre-extracted details for that file from **GET/uploads/{fileId}/extracted**. These details are the structured rows that were produced earlier by the extraction process (document type, invoice number, supplier, date, amount, and signature), so the Admin can verify the content without running any additional processing. The frontend simply renders what the backend already stored.

After reviewing, the Admin takes one of two actions. If the document is valid, the app sends **POST/admin/uploads/{fileId}/approve**. If it must be rejected, it sends **POST /admin/uploads/{fileId}/reject** with an optional reason. The Flask API authenticates the caller, confirms the Admin role, and then updates the database: it changes the file's upload_status to either "approved" or "rejected," records verified_by_user_id to capture who

made the decision, and stamps `verified_at` with the decision time. For rejections, it also stores `reject_reason`.

Once the backend confirms success, the frontend refreshes the pending list so the decided file disappears from the queue. Because every request carries a Bearer token and the backend enforces role checks on the admin endpoints, only authorized Admins can view pending files, read extracted details, and change a document's status.